

Results

01 • WELCH'S ANOVA

3+ groups, unequal variances

Table 1. Descriptives by group

Group	N	M	SD
control	20	32.48	7.47
drug_a	20	33.84	6.91
drug_b	20	37.81	7.79

Table 2. Welch's ANOVA

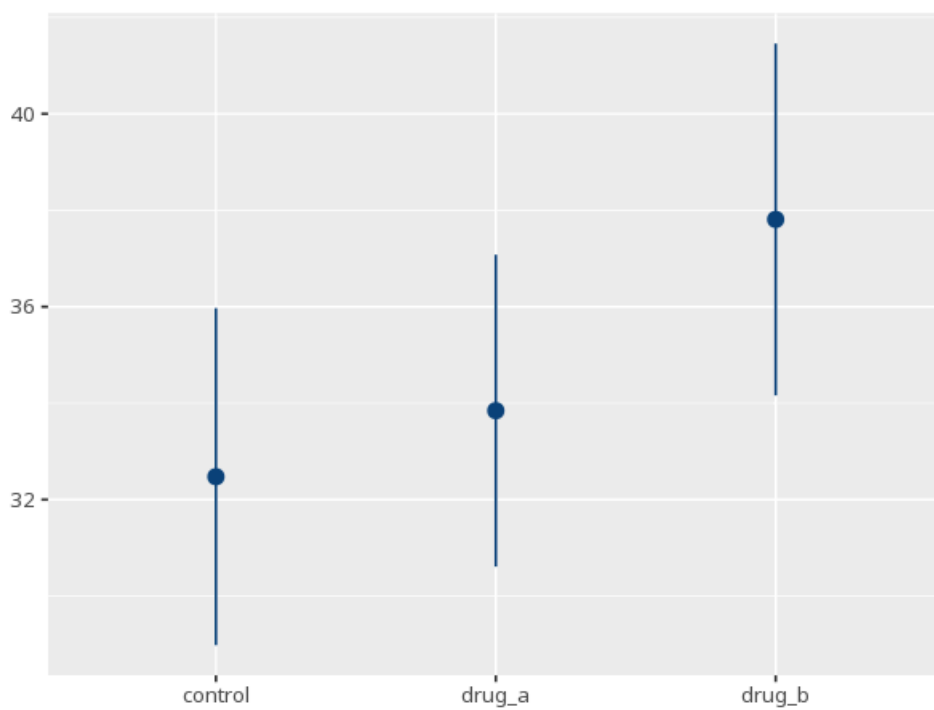
F	df1	df2	p
2.58	2	37.90	.089

Table 3. Games-Howell post-hoc

Pair	M _{diff}	p _{adj}	95% CI
control - drug_a	-1.37	.820	[-6.92, 4.18]
control - drug_b	-5.33	.082	[-11.22, 0.55]
drug_a - drug_b	-3.96	.218	[-9.65, 1.72]

Welch's adjusts the degrees of freedom so equal variances are not assumed (df2 is fractional); within-group normality is still assumed and checked with Shapiro-Wilk per group. (Shapiro per group: control W=0.97, p=.688; drug_a W=0.90, p=.050; drug_b W=0.92, p=.113)

Figure. Group means



How to read this test

A one-way ANOVA that relaxes the equal-variance assumption but still assumes roughly normal data within each group. A significant F means at least one group mean differs from the others; the Games-Howell post-hoc (also variance-robust) then shows which specific pairs differ. Your significance threshold (α) is 0.05.

APA template: Welch's ANOVA gave $F(2,37.90)=2.58$, $p = .089$.

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